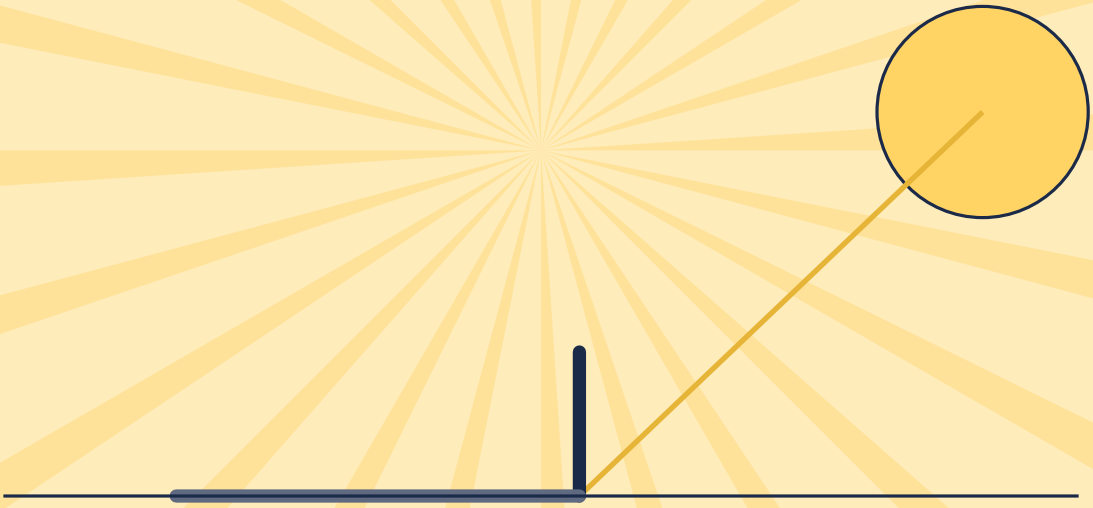
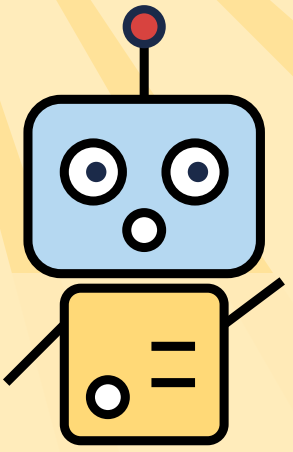
SCIENCE
Book 3

Shadows That Tell Time



Trace it. Measure it. Build a clock from a stick.

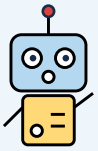
Puzzles, projects and Multix the guide robot



Ages 9 to 11

This book belongs to _____

Hello. I am Multix.

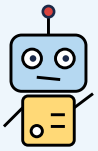


I am a guide robot. I read every instruction very carefully, which is why I sometimes get things wrong. When I make a mistake, see whether you can catch it. I never give away a puzzle answer.

How this book works

Every book in this series asks the same few questions, so once you have read this page you will know your way around.

- **Try it** boxes tell you what to do with a ruler, a cup, some string or a clock. You need nothing from a shop. Everything is at home or at school.
- **What did we keep?** After you simplify something, such as keeping the same conditions in a test or averaging several throws, you say what stayed and what was set aside. Simplifying is fine. Forgetting what you set aside is not.
- **What makes this correction trustworthy?** When you fix or improve an answer, a box says what you checked, what the check gives you reason to trust, and what it does *not* show.
- **Circles** show how a number was made. Page 4 explains how to read them, and Book 1 introduced them.
- **Puzzles** come at the end of each chapter. Try them before you look at the answers on page 44.
- **Slop check** is a page near the end. Slop is hasty, gaudy work that forgot to take care. You will learn to spot it and to fix it.



Here is a fact about me. I held my hand in front of a torch and the shadow on the wall was huge. I decided that my hand had grown. Chapter 2 will explain why I should be embarrassed.

In this book

1	Light Goes Straight	5
2	Big Shadow, Small Shadow	10
3	A Stick in the Sun	15
4	Building a Shadow Clock	22
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⋆	Slop check	37
⋆	Answers	44
⋆	Words, and notes for grown-ups	47

Try it: Gather your kit

You will use these things across the six chapters. Everything is found at home or at school. Measuring comes from Book 1 and fair testing from Book 2.

- a torch (a phone torch is fine)
- a stick or ruler, about 30 cm
- a ruler and a tape measure
- cardboard, scissors and tape
- chalk or small stones
- a small mirror
- a sunny day or three
- a watch or clock
- a pencil and a notebook
- a calculator (optional)

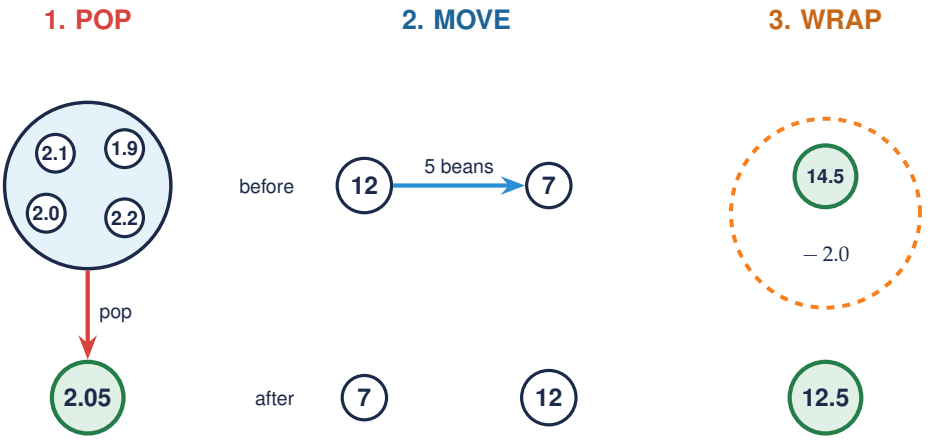
■ **Adult help:** **Never look at the sun**, not even for a second, and never aim a mirror at anyone's eyes. In this book you will study the sun only through the shadows it makes. Scissors and tape need care. Nothing here needs a flame or mains electricity.

Units in this book

We use the metric system: centimetres (cm), metres (m), minutes (min) and hours (h), and degrees for angles. There are 10 mm in 1 cm and 100 cm in 1 m. If you are used to inches, one inch is about 2.5 cm. We will use that only in side notes.

Reading the circles

A circle holds a group of things together. There are three things you can do with circles in this series.

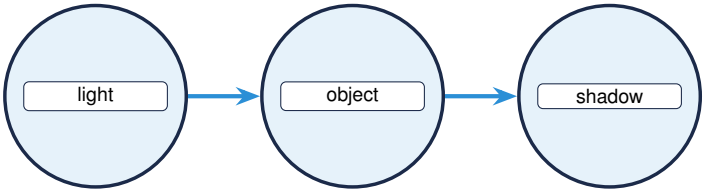


Pop Squash a group down to one value that sums it up, such as an average. A pop is a *summary*, so it always loses something. Every time we pop, we say what the summary keeps and what it loses.

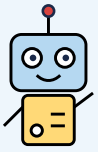
Move An arrow that carries something from one circle to another, such as beans or water. Nothing is summarized. You will see more of moves in later books.

Wrap When you correct something, draw a new dashed circle around the old one and show what changed. The old work stays visible, so anyone can check it.

A shadow is three circles in a line



Move the light, and the shadow circle changes. One change, one follow-up.



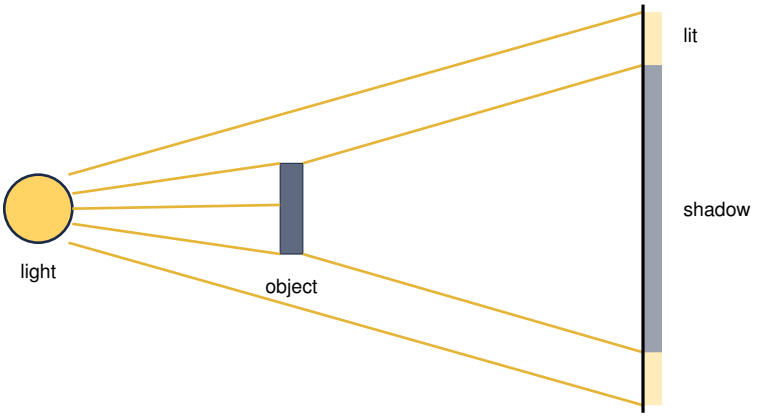
A pop is not a lie, but it is a shortcut. Any time I tell you a pop, I will tell you what it threw away.

1 Light Goes Straight

Why shadows have the shapes they have

Stand in front of a lamp and hold up your hand. A dark patch appears on the wall behind it. That patch is a **shadow**. It is not a thing. It is a place where something blocked the light.

Shadows work because of one simple fact: **light travels in straight lines**. When a straight line of light meets an object, the object stops it. Behind the object, that line never arrives, so the wall stays dark there.



Try it: Is light really straight? Three cards and a torch

You need: three pieces of card, a pencil to make a hole in each, a torch, and some putty or books to stand the cards up.

1. Make one small hole in the middle of each card, all at the same height from the bottom edge.
2. Stand the cards in a row on a table, about 20 cm apart, with the holes lined up. Shine a torch through the first hole from a few centimetres away.
3. Look at the third card. Is there a bright dot on it?
4. Slide the middle card a few centimetres sideways. What happens to the dot?

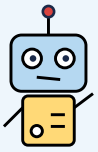
Write what you saw: _____

■ **Adult help:** Use a torch, not the sun, and never look into a bright beam. Pencils make holes in card best with something soft underneath.

What the cards show

When the three holes lined up on a straight line, light got through. When you moved one card, the dot disappeared. If light could bend around corners, it would have gone

around the moved card. It did not. That is evidence that light travels in straight lines, at least through the air in a room.



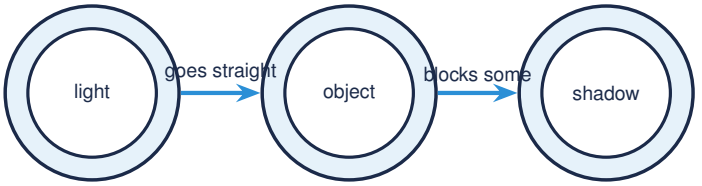
I have a plan. I will look around a corner by staring really hard. If light goes straight, then my eyes are very strong, so the corner should not matter. Is that a good plan?

Your eye needs light to arrive

You see an object when light from it reaches your eye. A shadow is a place where light did not reach. So when you look at your own shadow on the ground, you are looking at the ground where your body stopped the sun.

Three things in every shadow

Every shadow has three parts. You can draw each one as a circle, and an arrow shows what the light does next.



kept: where the light is and where the object is
set aside: colour and brightness

When you change the light (move it, raise it, lower it), the shadow circle changes with it. Next chapters use that idea to predict shadows before you see them.

What did we keep?

Drawing light as a straight line is a simplification. **We kept** _____ **and we set aside** _____.

Shadows have edges

A shadow's edge is where a straight line of light just misses the object. If the light is a tiny point, such as a small torch bulb far away, the edge is sharp. If the light is wide, such as a big window, light comes at the object from many places, and the edge looks blurry. We will draw the sharp kind in this book.

Try it: Sharp or blurry?

In a dim room, hold a pencil 5 cm in front of a white wall. Shine a small torch from far away, then a bigger lamp from close. Which makes the sharper shadow edge?

_____

Which light was closer to being a single point? _____

Puzzle 1.1: Straight lines

Three pencils stand in a row on a table, all the same height. A torch on the table shines along the row from the left. A shadow falls on the wall behind.

(a) How many separate shadows would you see if the torch is in line with the pencils?

_____

(b) Why? _____

Puzzle 1.2: Is it a shadow?

Write S if it is a shadow (light blocked) and N if it is not.

(a) The dark patch under a tree on a sunny day. _____

(b) The reflection of a tree in a pond. _____

(c) The dark shape of your head on a wall in front of a lamp. _____

(d) A painting of a night sky. _____

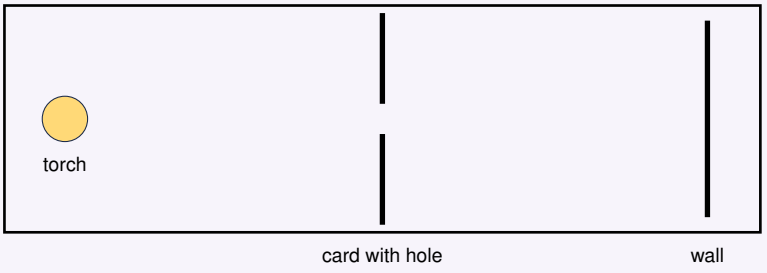
Puzzle 1.3: Trace the line

A torch, a card with a small hole and a wall are in a line. The torch is 40 cm from the card and the card is 40 cm from the wall.

(a) Draw the straight line of light from the torch through the hole to the wall.

(b) If you move the card 5 cm up, does the dot on the wall move up, down or stay?

_____

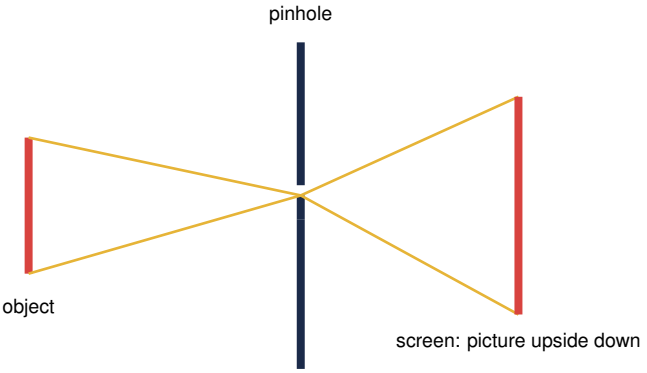


Puzzle 1.4: Corner

A ball sits behind a wall. You stand where you cannot see it. Draw (or say) why you cannot see it, using lines of light.

A pinhole picture

Straight lines of light can make a picture. Poke a tiny hole in a card and hold it between a bright lamp and a white wall, in a dim room. A faint, upside-down picture of the lamp appears on the wall. Light from the top of the lamp travels in a straight line through the hole and lands on the lower part of the wall. Light from the bottom of the lamp lands on the upper part. The lines cross at the hole, so the picture is flipped.



Try it: Make a pinhole picture

You need: a shoebox or a card tube, scrap of baking paper, tape, a pin, a bright lamp (not the sun).

1. Cut a small square window in one end of the box and tape baking paper over it as a screen.
2. Poke one pin hole in the middle of the opposite end.
3. Point the hole at a lamp in a dim room. Look at the baking paper from behind the box. Describe the picture. Is it upside down?

_____

■ **Adult help:** Never point a pinhole box at the sun and look through it. Use a lamp.

Predicting the picture (practice numbers)

A candle-sized lamp 10 cm tall is 40 cm in front of the pinhole and the screen is 10 cm behind it. The picture's height is $10 \times 10 \div 40 = 2.5$ cm. A practice measurement gave 2.6 cm. That is 0.1 cm from the prediction.

Puzzle 1.5: Upside down

(a) A flame is 8 cm tall, 40 cm in front of a pinhole, and the screen is 10 cm behind. How tall is its picture? _____ cm

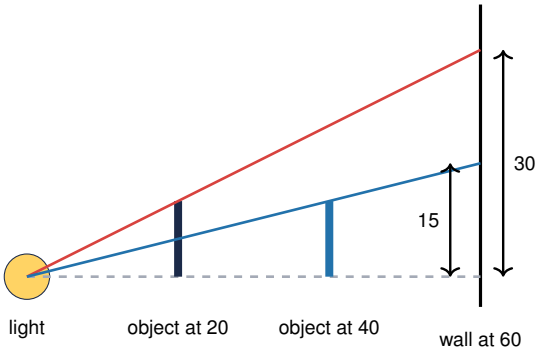
(b) If you move the flame to 20 cm from the hole, is the picture bigger or smaller? _____

2 Big Shadow, Small Shadow

What decides how large a shadow is

Hold your hand close to a torch and the shadow on the wall is enormous. Pull your hand away and it shrinks. Your hand did not grow or shrink. The *shadow* did.

Light travels in straight lines, so you can predict the size of a shadow by drawing the lines. Draw a line from the light past the top of the object. Where it hits the wall is the top of the shadow.



The red line passes the top of the object at 20 cm from the light. It reaches the wall at 30 cm up. The blue line passes the top of the same object at 40 cm from the light. It reaches the wall at only 15 cm. The closer the object is to the light, the bigger its shadow.

Try it: Shadow puppet test

Question: **does moving a puppet closer to the torch make its shadow taller?** Use the fair-test habits from Book 2.

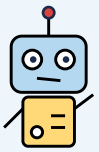
1. Cut a card puppet 10 cm tall. Tape it to a stick so it stands up.
2. Put a torch on a stack of books so it sits at the puppet's height. Tape a white sheet to the wall, 60 cm from the torch.
3. Hold the puppet 20 cm from the torch, then 30, 40 and 50 cm. Keep the torch and the wall where they are.
4. At each distance, measure the shadow's height on the wall, to the nearest cm. Do each distance twice.

Distance from light (cm)	Shadow, try 1 (cm)	Shadow, try 2 (cm)	Average (cm)
20			
30			
40			
50			

■ **Adult help:** Use a torch or phone light. Do not use a hot bulb that someone could touch.

What stays the same, and what changes

I change	how far the puppet is from the light: 20, 30, 40 or 50 cm
I measure	the height of the shadow on the wall, nearest cm
I keep the same	the puppet (10 cm tall), the torch, the wall's distance (60 cm), the torch height



I have a plan. I will move the puppet closer to the light *and* move the wall further away, to get an even bigger shadow. Then I will say the puppet position made it bigger. Is that a fair test?

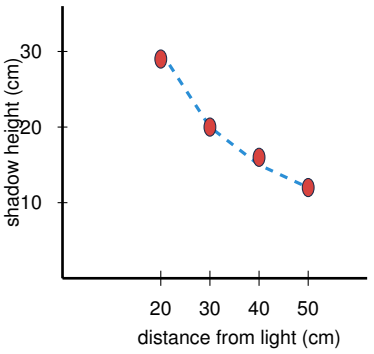
Drawing predicts, measuring checks

Before you measure, you can predict by drawing. The drawing above says the shadow should be 30, 20, 15 and 12 cm at distances 20, 30, 40 and 50 cm. Here are practice numbers from a made-up run.

Shadow puppet results (practice numbers, not from a real test)

Distance (cm)	Drawing says (cm)	Measured (cm)	Off by (cm)
20	30	29	1
30	20	20	0
40	15	16	1
50	12	12	0

The drawing was within 1 cm each time. That is a good sign for the drawing.



The dashed line is the drawing's prediction and the dots are the measurements. They sit almost on top of each other.

What makes this correction trustworthy?

What we checked. We drew a prediction first, then measured the shadow heights, changing only the puppet's distance from the light and doing each distance twice.

What this check gives us reason to trust. When a drawing made from straight lines predicts the measured heights to within 1 cm at four distances, the straight-line idea is working. Changing only the distance means the distance, not the torch or wall, is the likely reason for the change.

What it does not show. That the drawing works for fuzzy lights, curved objects or very big shadows. It also does not show *why* light goes straight. Four distances and one puppet are a small test.

A pattern in the numbers

Look at the last column of the table in the worked box: the shadow gets smaller as the puppet moves away, but not by the same amount each time. From 20 to 30 cm it drops by 10 cm, from 30 to 40 it drops by 5, and from 40 to 50 it drops by 3. Equal steps in distance do not give equal steps in shadow height.

What did we keep?

We predicted shadows with straight-line drawings. **We kept** _____ **and we set aside** _____.

Puzzle 2.1: Predict the shadow

Use the drawing method: the object is 10 cm tall and the wall is 60 cm from the light.
(a) The object is 15 cm from the light. How tall is the shadow? _____ cm
(b) The object is 60 cm from the light, right against the wall. How tall is the shadow? _____ cm
Hint: shadow height $= 10 \times 60 \div$ distance.

Puzzle 2.2: Spot the unfair test

Mara tests how the distance from the light changes a shadow. She moves the puppet closer to the torch, and she also moves the torch to a lower shelf. The shadow gets taller.
(a) What did Mara mean to change? _____
(b) What else changed? _____
(c) How would you fix the test? _____

Puzzle 2.3: Check the numbers

A student measured these shadows for a 10 cm puppet at 20, 30, 40, 50 cm: 30, 20, 15, 11 cm.
(a) What does the drawing predict for 50 cm? _____
(b) How far off was the 11 cm reading? _____
(c) Does this change your trust in the drawing? Why or why not?
_____

Puzzle 2.4: Bigger or smaller?

Write B (bigger) or S (smaller) for the shadow:
(a) You move your hand toward the torch. _____
(b) You move the wall closer to the torch and the hand stays still. _____
(c) You use a taller puppet at the same distance. _____

More shadow sums

Use the same rule: shadow height = object height × wall distance ÷ object distance.

Puzzle 2.5: Your hand

Your hand is 18 cm tall. The wall is 60 cm from the torch.
(a) The hand is 20 cm from the torch. Shadow: _____ cm
(b) The hand is 40 cm from the torch. Shadow: _____ cm

Puzzle 2.6: Two puppets

A 10 cm puppet and a 20 cm puppet are both 30 cm from the torch, wall at 60 cm.
Shadows: _____ cm and _____ cm. How many times taller is the second shadow?
_____

Puzzle 2.7: Move the wall

A 10 cm puppet is 20 cm from the torch. The wall is moved from 60 cm away to 80 cm away.
New shadow height: _____ cm. Did it grow or shrink? _____

Puzzle 2.8: Read the graph

From the graph earlier in this chapter, about how tall was the shadow at 30 cm? _____ cm
At what distance was it about 15 cm? _____ cm

Puzzle 2.9: Explain to a friend

In one or two sentences, explain why the shadow of your hand grows when you bring it closer to the torch.

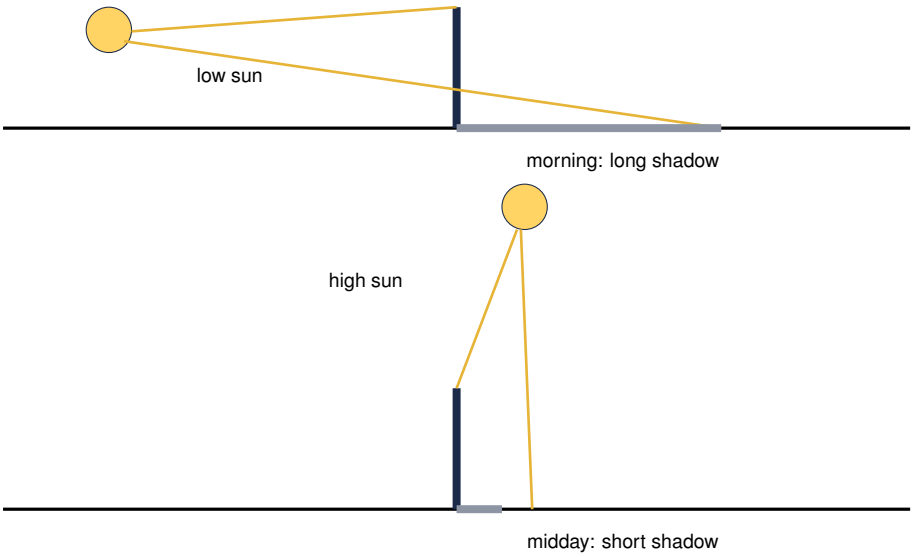
_____

_____

3 A Stick in the Sun

Following the sun by what it leaves on the ground

The sun is a light, a very big one that is very far away. Because it is so far away, its light arrives almost in parallel lines, and everything on the ground gets the same direction of light. A shadow from the sun is long or short, and points one way or another, depending on where the sun is in the sky.



When the sun is low, light comes in at a shallow angle and the shadow stretches out. When the sun is high, the light comes steeply down and the shadow is short. A shadow also always points **away** from the sun. If the sun is in the east, the shadow points west.

■ **Adult help: Never look at the sun.** You will study the sun only by looking at the shadows on the ground. Do the tracing outdoors on a flat, safe spot, and take a hat and water.

Try it: Trace a stick's shadow all day

You need: a stick or ruler about 30 cm tall, modelling clay to hold it up, a large sheet of paper taped to a flat spot in the sun, a pencil and a clock.

1. Tape the paper down so it cannot move. Stand the stick upright in clay on the paper and trace round the base. Do not move the stick or the paper again.
2. On the hour, mark the tip of the shadow with a dot and write the time next to it.
3. Do this at least five times through the day, for example 9, 11, 1, 3 and 5.
4. Afterwards, draw a line from the base to each dot and measure each line.

Time						
Shadow length (cm)						

Fair test, again

Your paper and stick stay in one place. The only thing that changes is the **time**. Everything else, such as the stick's height and where the shadow falls, is kept the same, so any change in the shadow comes from the sun's movement.

Practice numbers

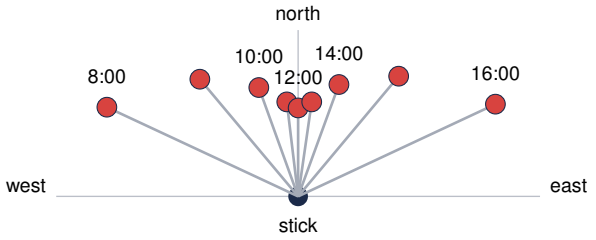
Here is a made-up day for a stick 30 cm tall, in a place in the northern half of the world. The numbers are practice numbers, not from a real day.

One practice day, a 30 cm stick

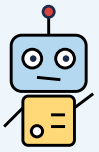
Time	8:00	9:00	10:00	11:00	12:00	13:00	14:00	15:00	16:00
Length (cm)	62	45	34	28	26	28	35	46	64

The shortest shadow is 26 cm at 12:00. The shadow is longest early and late: 62 cm at 8:00 and 64 cm at 16:00.

Plot the shadow tips, looking down from above. The stick stands at the dot at the bottom, north is up.



In the morning the shadow points west. At midday it points north, because in the northern half of the world the midday sun is in the south. By afternoon it points east. (South of the equator the midday shadow points south instead.) The tips draw a gentle curve.



The shadow is shortest at 12:00. So the sun must be highest at exactly 12:00 for everybody in the world. Is that right?

Shortest shadow is not always at 12:00

The sun is highest in the sky at a time called **solar noon**. Solar noon depends on where you live, and on whether your clocks have been moved for summer. In many places in summer it is closer to 1:00 on the clock. The practice day above happens to have its shortest shadow at 12:00. Your day may be different. The shortest shadow is the best sign of solar noon, not the clock.

Try it: Two sticks, one ratio

At the same time, measure the shadows of two sticks of different heights, standing on the same flat ground.

	Stick height (cm)	Shadow (cm)	Shadow ÷ height
Short stick			
Tall stick			

Two sticks (practice numbers)

At 10:00, a 15 cm stick had a 17 cm shadow, a 30 cm stick had 34 cm and a 45 cm stick had 51 cm. $17 \div 15 = 1.13$, $34 \div 30 = 1.13$, $51 \div 45 = 1.13$. The ratio is the same for all three, about 1.1.

At a given time, every upright thing makes a shadow that is the same number of times as long as the thing is tall. Taller things make longer shadows, in the same proportion.

What makes this correction trustworthy?

What we checked. We traced the shadow tips of one stick through a day on paper that did not move, and we checked with two more sticks at one time.

What this check gives us reason to trust. Keeping the stick and paper fixed means any change in the shadow comes from the sun's changing position. Three sticks giving the same ratio shows the idea of straight, parallel sunlight is working.

What it does not show. Why the sun moves, or what the shadows would look like on another day of the year. One day, one place and a few sticks are a small test.

What did we keep?

The sun is reduced to a direction and a height. **We kept** _____ **and we set aside** _____.

Puzzle 3.1: Read the table

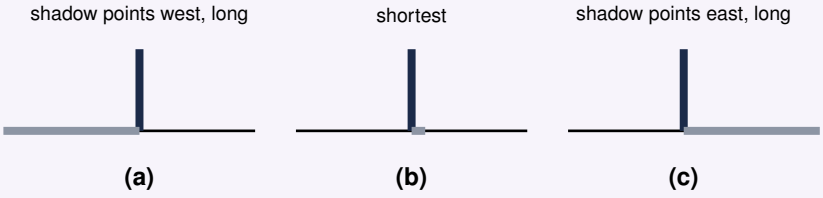
Use the practice day: 8:00 62, 9:00 45, 10:00 34, 11:00 28, 12:00 26, 13:00 28, 14:00 35, 15:00 46, 16:00 64 (cm).
(a) When is the shadow shortest? _____
(b) How much longer is the 9:00 shadow than the 12:00 shadow? _____ cm
(c) Which is longer, the 10:00 or the 14:00 shadow? _____

Puzzle 3.2: Which way?

The sun is in the east. Which way does a shadow point? _____
At 3:00 in the afternoon in Canada, does the shadow point west, north or east? _____

Puzzle 3.3: Match the shadow

Write M (morning), N (midday) or A (afternoon) for each stick.



(a) _____ (b) _____ (c) _____

Puzzle 3.4: Where is the sun?

A shadow points toward the top of the page (north). Where is the sun, north, south, east or west? _____
Is the shadow long or short if the sun is high? _____

Puzzle 3.5: Check the ratio

At noon a 20 cm stick has a 22 cm shadow. How long should the shadow of a 40 cm stick be at the same time? _____ cm
How long should a 10 cm stick's shadow be? _____ cm

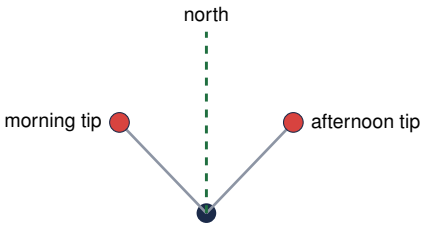
Finding north with a stick

Your traced tips give you a way to find north. In the morning and afternoon, there is a time when the shadow has the *same length* as it did earlier. The line that splits the two shadows in half points north (in the northern half of the world).

Try it: Find north

Use your traced page. Pick a morning tip and an afternoon tip with the same shadow length. Draw the two lines from the stick to those tips. Draw a line that splits the angle between them in half. That line points north. Check it with a map or a phone compass.

Morning tip length _____ cm Afternoon tip length _____ cm



Practice numbers

On the practice day the 10:00 shadow is 34 cm and the 14:00 shadow is 35 cm. They are almost the same, so the line halfway between them points north. A difference of 1 cm is within what you would expect from measuring.

Puzzle 3.6: Almost the same

The 9:00 shadow is 45 cm and the 15:00 shadow is 46 cm. Can you use these two to find north? _____ Why? _____

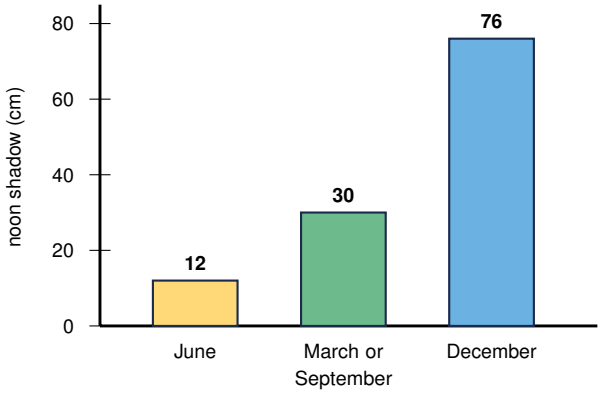
Puzzle 3.7: A lopsided day

A stick has a 50 cm shadow at 9:00 and a 30 cm shadow at 3:00. Can you use these two to find north? _____ Why or why not?

_____

Noon shadows through the year

Your stick makes a different shortest shadow in different seasons. In summer the sun climbs high, so the noon shadow is short. In winter the sun stays low, so the noon shadow is long. The numbers below are calculated, not measured, for a 30 cm stick at a place 45 degrees north of the equator, such as the Ottawa area, on three days.



The Earth's axis is tilted, so in June our half of the world leans toward the sun and in December away from it. That is why a shadow clock marked in one season drifts in another. The practice day, with a noon shadow of 26 cm, was between March and June.

Puzzle 3.8: Which season?

A 30 cm stick has a 70 cm shadow at noon. Is it more likely June, September or December?

_____

Why? _____

Puzzle 3.9: Shadow ratios

In December the noon shadow is 76 cm and the stick is 30 cm. How many times as long as the stick is the shadow? _____ Round to the nearest whole number.

4 Building a Shadow Clock

Turning a stick and a sunny day into a way to tell the time

People used shadows to tell the time long before there were watches. Archaeologists have found shadow clocks from ancient Egypt that are thousands of years old, and sundials were used across the ancient world. The idea is simple. If the shadow of a fixed stick moves in the same way every day, then *where the shadow is* tells you *what time it is*.

In the last chapter you traced one day of shadow tips. That tracing is already most of a clock. You only need to keep it still and use it again.

Try it: Mark your clock

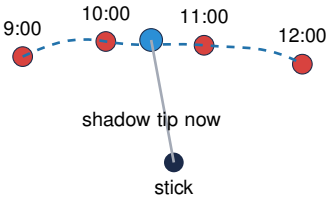
You need: the stick and the paper from Chapter 3, tape, and a pencil. A cardboard base makes it easier to keep everything in place.

1. On a sunny day, tape your paper to a flat board and put the board on a flat spot. Mark the board's position on the ground with chalk or tape.
2. Stand the stick upright in clay in the same spot on the paper. Trace round the base.
3. Each hour, mark the tip of the shadow with a dot and write the time. Write the *hour on the clock*, such as "10:00".
4. Later, join your dots with a gentle curve. That curve is your clock face.

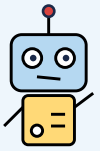
■ **Adult help:** The paper and board should be held flat with tape and the stick must be steady. Put the clay stick where no one will trip over it.

Reading the clock

On a later day, put the board back in exactly the same place, with the stick at the traced base. Wait for the shadow tip to reach the curve. The nearest dot tells the time. A tip between two dots means between those two hours.



The sky-blue dot is the tip today. It is about two thirds of the way from 10:00 to 11:00, so you would say **about 10:40**.



I have a plan. I will draw my clock face in the evening, in the dark, using my torch. The hours will be exactly where I want them. Does that make my clock accurate?

Checking against a real clock

A clock is only worth anything if it agrees with a real one. Check yours. On a later day, look at the shadow and a watch together. Write what the shadow clock says, what the watch says, and how far apart they are.

Try it: Check your clock on more than one day

Pick three times when the shadow is on or near a dot. Compare with a watch. Repeat a few days later.

Shadow clock says	Watch says	Off by (min)	Day

Checking a shadow clock (practice numbers)

Day 3 after marking. The shadow said 10:00 when the watch said 10:04, 12:00 when it said 12:05, 14:00 when it said 14:03. Off by 4, 5 and 3 min. Average error $(4 + 5 + 3) \div 3 = 4$ min, spread 2 min.

Day 14. The shadow said 10:00 when the watch said 10:12, 12:00 when it said 12:14, 14:00 when it said 14:11. Off by 12, 14 and 11 min. Average error $(12 + 14 + 11) \div 3 \approx 12$ min, spread 3 min.

The clock is within about 4 minutes just after it was made, and about 12 minutes after two weeks. The gap between the two errors, 8 minutes, is bigger than either spread (2 and 3). That fits what we expect: the sun's path through the sky changes slowly through the year, so a clock marked on one day slowly drifts. Real sundials are built to avoid this by tilting the stick, which is called the **gnomon**, to point at the pole star.

What makes this correction trustworthy?

What we checked. We marked the clock on one day and compared it with a real watch on a later day and again two weeks later.

What this check gives us reason to trust. Each comparison is against something independent, a watch, so errors cannot hide. The errors were small just after marking and larger later, which fits what we know about the sun's changing path.

What it does not show. Whether the clock works in another season, on a cloudy day, or at another place. Two checking days and three times each are a small test.

What did we keep?

A shadow clock reduces the day to marked dots. **We kept** _____ **and we set aside** _____.

Why a clock marked once drifts

In summer the sun climbs higher than in winter, so the tip of the shadow for a given hour lands in a different place on a different week. A clock marked on one day matches that day best. A real sundial gets around this with a tilted stick, the gnomon, which points at the pole star so that the shadow's direction depends only on the hour.

Try it: Mark it twice

Mark your clock's hour dots in pencil on one day. A week or two later, mark the tips again for the same hours in a different colour. Do the two sets of dots match? How far apart are they for 10:00, 12:00 and 2:00?

10:00 _____ cm 12:00 _____ cm 2:00 _____ cm

Puzzle 4.1: Read the clock

The dots are 9:00, 10:00, 11:00 and 12:00. The shadow tip is halfway between 11:00 and 12:00. What time is it? _____

A friend says it is 10:00 because the shadow is long. Is the friend right? Explain.

_____

Puzzle 4.2: How far off?

A shadow clock said 11:00, 1:00 and 3:00 when the watch said 11:06, 1:09 and 3:03.

(a) Errors: _____ _____ _____ min

(b) Average error: _____ min (c) Spread: _____ min

Puzzle 4.3: Drift

Day 3 errors: 4, 5, 3 min. Day 14 errors: 12, 14, 11 min.

(a) Gap between averages: _____ min.

(b) Is the gap bigger than both spreads? _____

(c) What does this suggest about the clock? _____

Puzzle 4.4: Plan a check

Fill in a card for checking a shadow clock fairly.

Question _____

I change _____

I measure _____

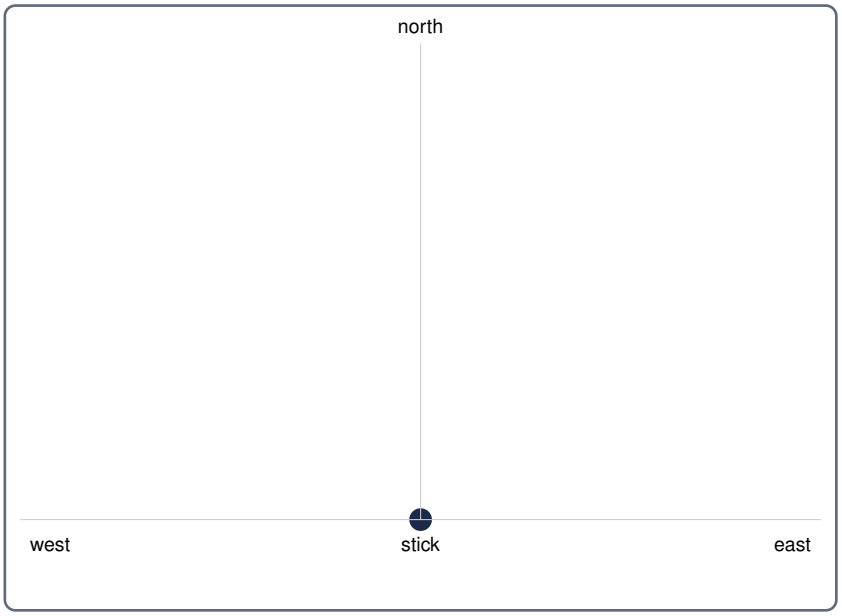
I keep the same _____

Puzzle 4.5: Cloudy day

On a cloudy day the shadow clock stops working. Why? _____
Name one way people could still tell the time then. _____

My shadow clock record

Use this page to draw your own clock face from a real sunny day. Put a dot at the stick's base, then at each hour mark the tip of the shadow. Write the hour next to the dot.



Date: _____ Place: _____ Stick height: _____ cm

Shortest shadow at: _____ o'clock Length: _____ cm

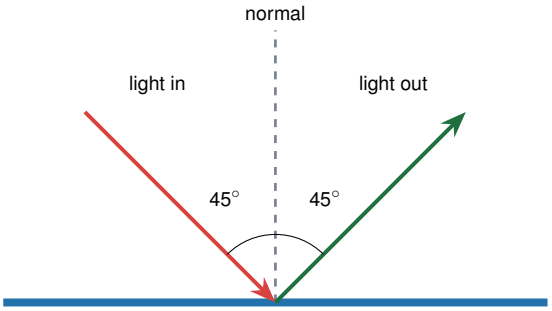
5 Mirrors and Light Paths

Where a beam goes when it hits something shiny

So far light has gone straight. A mirror changes that. When light hits a mirror it bounces off, and it bounces in a predictable way. Once you know the rule, you can trace a beam through a whole maze of mirrors, and you can predict where it will land.

The rule of the bounce

Draw a line that sticks straight out from the mirror at the spot where the light hits. This line is called the **normal**. The angle between the incoming light and the normal equals the angle between the outgoing light and the normal. Same angle in, same angle out.



Try it: Bounce a torch beam

You need: a small flat mirror, a torch, a sheet of paper, a ruler, a pencil, a protractor if you have one.

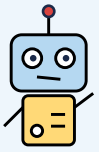
1. Tape the paper on a table. Draw a straight line near the middle and stand the mirror along it. Draw the normal at one point.
2. Shine the torch along the table from one side so the beam hits that point. Mark two dots on the beam going in, and two dots on the beam coming out.
3. Remove the mirror, join the dots, and measure the angle in and the angle out.

Angle in: _____° Angle out: _____°

■ **Adult help:** Keep the torch low and along the table. Never shine a mirror toward anyone's eyes, and never use sunlight on a mirror. A mirror can send bright light a long way.

Bounce results (practice numbers)

Three tries, angle in against angle out: 30 and 29, 45 and 46, 60 and 58 degrees. The differences are 1, 1 and 2 degrees. Each is small enough to put down to ruler and protractor error.



My angle in was 30 degrees. My angle out was 60. The rule says they should match. Obviously the rule is wrong and my protractor is perfect. Do you agree?

What makes this correction trustworthy?

What we checked. We measured the angle in and the angle out for three bounces and compared them.

What this check gives us reason to trust. Three different angles agreeing to within 2 degrees supports the rule that the angle in equals the angle out. The differences are about the size of the errors a ruler and protractor make.

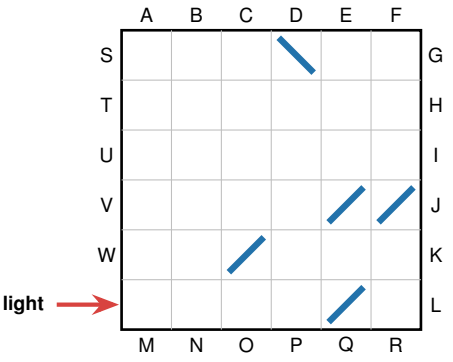
What it does not show. That the rule holds for bumpy surfaces, curved mirrors or very steep beams. Three bounces with one mirror are a small test.

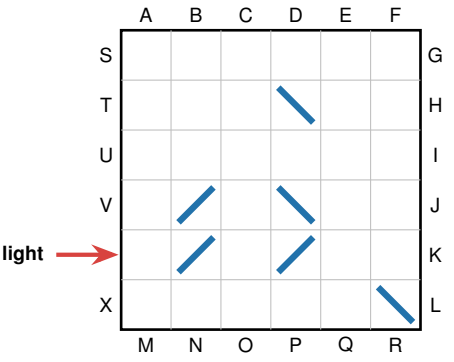
What did we keep?

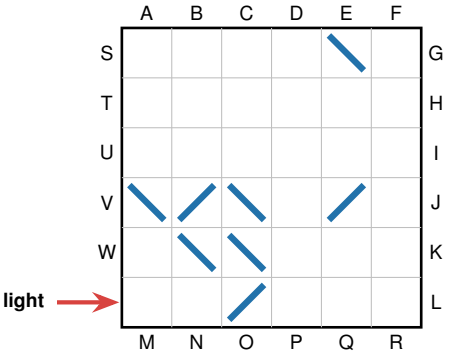
We drew light as a thin line with angles. **We kept** _____ **and we set aside** _____.

A maze of mirrors

In the mazes below, each mirror sits on a diagonal of a square. A beam that travels along a row, column or any line of squares turns 90 degrees at a mirror. A beam going east that hits a slanting mirror that rises to the right (/) turns north. One that hits a mirror that falls to the right (\) turns south.







Puzzle 5.1: Trace the beam

Light enters from the left at each arrow and travels east. Trace it with a coloured pencil. Which letter does it leave at?

Maze 1: _____ Maze 2: _____ Maze 3: _____

Puzzle 5.2: The bend

Light going *north* hits a (/) mirror. Which way does it go? _____

Light going *west* hits a (\) mirror. Which way does it go? _____

Puzzle 5.3: Periscope

A periscope uses two mirrors to let you see over a wall. Light from a tree travels west, hits a mirror, turns up, hits a second mirror and turns west again. Draw the two mirrors and the beam.

_____

Puzzle 5.4: Check the rule

A beam hits a mirror at 40° to the normal. At what angle does it leave? _____°
A student measures 36° going out. How far off is this? _____° Is that likely just measuring error? _____

Mirror writing

Hold a mirror beside a word. The mirror swaps left and right but not up and down. Letters such as A, H, I, M, O, T, U, V, W, X and Y look the same in a mirror held to the side. Letters like B, D, F, G, R and S do not.

Try it: Mirror name

Write your name on paper. Hold a mirror upright at the right edge of the word. Does it still read the same? Which letters changed?

Name: _____ Letters that changed: _____

Puzzle 5.5: Same in the mirror?

Which of these words look the same in a mirror held beside them? MOM DAD WOW TOOT HAT

Answer: _____

Puzzle 5.6: Mirror tricks

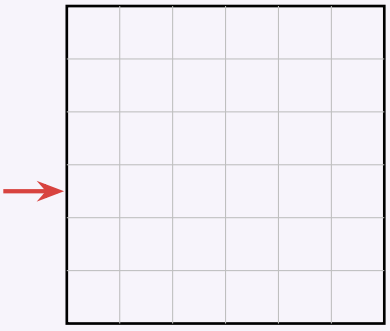
A mirror shows a clock reading 3:00 with the hour hand on the left. What does the real clock show? _____

Write one sentence about what else looks different in a mirror, such as your right hand.

_____

Puzzle 5.7: A better maze

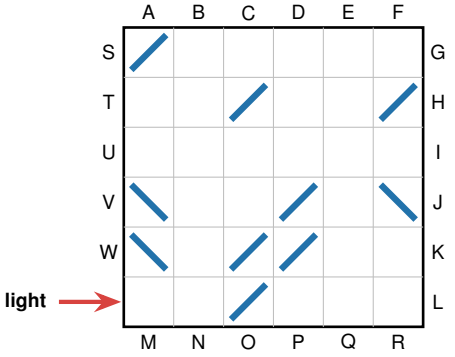
Design a small mirror maze of your own on the grid. Put three mirrors in and trace the beam.

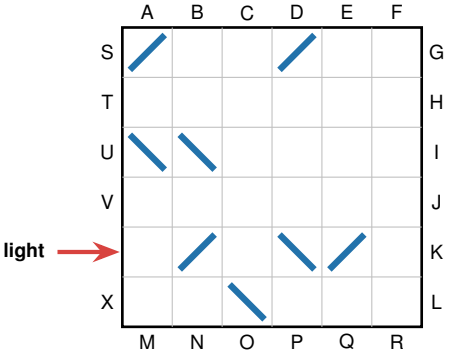


Where does your beam leave? _____

Mirror maze challenge

These mazes have more mirrors. Trace each beam with a coloured pencil, turning at every mirror.





Puzzle 5.8: Harder mazes

Maze A (left): the beam leaves at _____. Maze B (right): the beam leaves at _____.
How many mirrors did each beam bounce off? A: _____ B: _____

Puzzle 5.9: Reverse the beam

If light is sent back in at the exit of a maze, will it follow the same path in reverse? _____ Test it on Maze 1 of this chapter.

_____

6 What a Shadow Keeps

Saying what the model keeps and drops

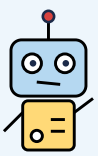
A shadow is a simple stand-in for a complicated thing. The shadow of a tree on the ground keeps the tree's outline and drops its colour, its texture and its leaves one by one. If you only had the shadow, you could still say a lot about the tree's height and the sun's position, but you could not say what colour the leaves were.

That is what a **model** does. A model keeps some things and drops others, on purpose. Every drawing in this book was a model of light.

Model	What it keeps	What it drops
A straight line for light	the direction light goes	the colour, the brightness, the wave
A shadow	the object's outline and where the light is	colour, texture, inside
A dot for the sun	its direction and height	its size and brightness
A shadow clock	the hour from the tip's place	minutes, the season
A mirror rule	angle in equals angle out	bumpy surfaces

Old measurements, reported

Over two thousand years ago a Greek scholar named Eratosthenes is reported to have compared shadows at two places in Egypt on the same day to estimate how big the Earth is. The account comes from a later writer, Cleomedes, so historians treat the details with care. The idea is that a shadow's angle is a measurement that holds information about where on the globe you are standing.



My shadow clock is a perfect model of the sun. It keeps everything about the sun. Is that right?

Five lines for a shadow report

The report from the earlier books fits shadows, too.

1. **What I looked at.** The stick, the light, the place and the day.
2. **What I changed, measured and kept the same.**
3. **My readings.** Times and lengths or angles, all of them.
4. **My answer.** The pattern, with averages or errors where they apply.
5. **What I kept and what it hides.**

A sample report (practice numbers)

1. Looked at. A 30 cm stick standing in clay on a taped sheet of paper, on a sunny day in the northern half of the world.
2. Changed, measured, kept. The time changed. I measured the shadow's length and direction. The stick, paper and place stayed the same.
3. Readings. Lengths at 8, 9, 10, 11, 12, 13, 14, 15 and 16:00: 62, 45, 34, 28, 26, 28, 35, 46 and 64 cm.
4. Answer. The shadow was shortest at 12:00, about 26 cm, and longest at the ends of the day. The shadow pointed west in the morning, north at midday and east in the afternoon.
5. Kept and hidden. I kept the shadow's length and direction. I set aside the sun's brightness, the day's weather and the season. A different day of the year may give different lengths.

Puzzle 6.1: Safe sentences

Mark each sentence S (supported by the shadow data), G (a guess about why) or B (too big for the data).
(a) "The shadow was 26 cm at 12:00." _____
(b) "Shadows are shortest at noon everywhere on Earth." _____
(c) "The shadow is short because the sun is high." _____

Puzzle 6.2: Model or real thing?

Write M if the sentence describes a model and R if it describes the real thing.
(a) A drawing of a beam as a straight line. _____
(b) The sun in the sky. _____
(c) A paper clock with hour dots. _____

Puzzle 6.3: What does it keep?

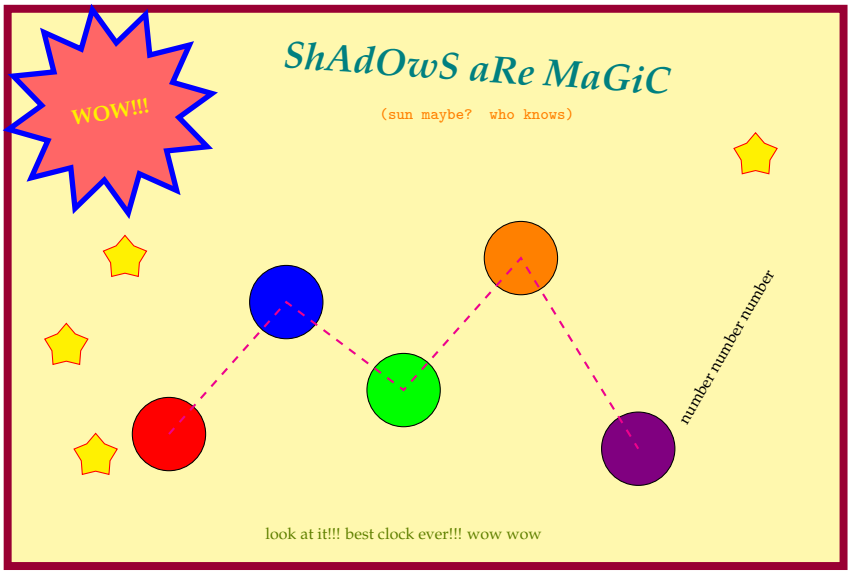
A shadow clock keeps the direction of the sun's shadow. Write one thing it drops.
_____

Slop check

Slop is hasty, gaudy work that forgot to take care. It can be made by a person or a machine, quickly or slowly. What matters is whether the work shows care for three things:

- **Salient:** Is the main point easy to see?
- **Simple:** Is there anything that does nothing but get in the way?
- **Relevant:** Does everything on the page help answer the question?

Here is a hasty poster about the shadow clock. Study it, then decide what is wrong.



Slop check

(a) What is the poster's main point? Can you tell?

_____

(b) Name two things that are not simple, because they only distract.

_____

(c) Name two things that are missing, so the dots cannot be read.

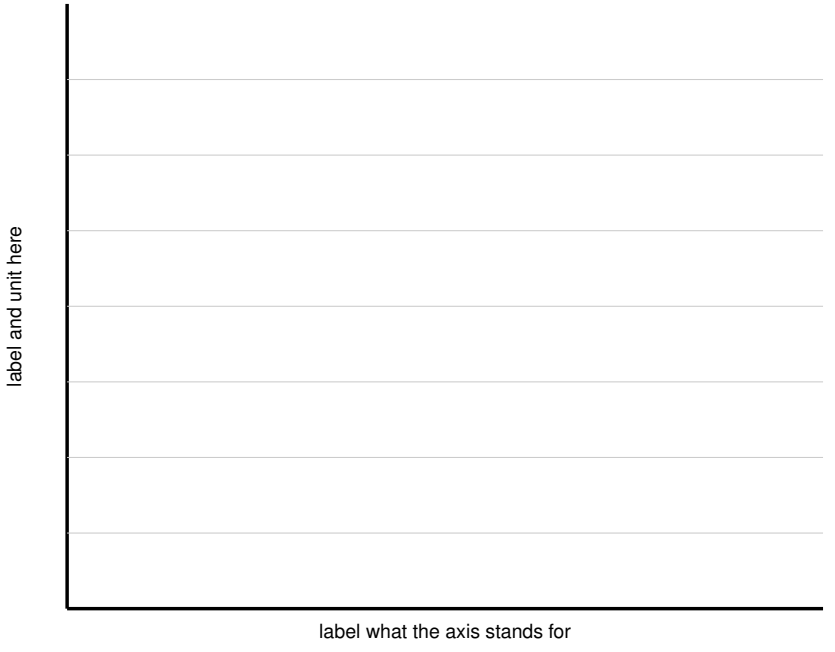
_____

(d) Which part is not relevant to the shadow clock at all?

_____

Now you try

On a separate sheet, redraw a clear picture of the practice day: shadow length against time, from 8:00 to 16:00, with the numbers 62, 45, 34, 28, 26, 28, 35, 46 and 64 cm. Give it a title, label the axes with units, use one or two calm colours, and delete everything that does not help. Check it against the three words: salient, simple, relevant.

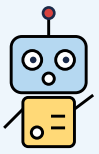


Title:

What did we keep, across the whole book?

For each chapter, say one thing that was simplified, what was kept, and what was set aside. A sample is done for you.

Chapter	What we simplified	What we kept	What we set aside
1 Light goes straight	light drawn as a line	the direction light goes	colour and brightness
2 Shadow size			
3 A stick in the sun			
4 A shadow clock			
5 Mirrors			
6 Shadow keeps			



Everything you traced in this book was a simplification of something. A model is allowed to drop things, as long as you say which. Next time you see a long evening shadow, you will know what it is telling you.

Mixed review

Puzzle 6.4: Which is which?

Write L (longest shadow), S (shortest shadow) or M (in between) for each time in the practice day: 8:00 _____ 10:00 _____ 12:00 _____ 14:00 _____ 16:00 _____

Puzzle 6.5: Test detective

(a) “I measured the shadow of a stick on a hill at 9:00 and the shadow of a stick on a flat road at 3:00 and found the second was longer.” What changed besides the time?

_____

(b) “I moved the lamp closer and the puppet away at once.” What is wrong?

_____

Puzzle 6.6: Putting it together

A 30 cm stick has a 34 cm shadow at 10:00. A building has a 340 cm shadow at the same time. How tall is the building? _____ cm Hint: use the ratio.

Puzzle 6.7: Short answer

Name one thing a shadow clock keeps and one it drops.

_____

Shadow projects to try

1. Measure the shadow of a tree or lamp post at three times. Use the ratio to work out its height.
2. Make a shadow puppet theatre and measure how big the puppets look from different distances.
3. Build a pinhole box and compare the picture size with the object size.
4. Draw the shadow of your house at 8:00, 12:00 and 16:00 on a sketch map.
5. Make a periscope from two small mirrors and a box.

Project	What I changed	What I measured	What I kept the same

My shadow diary

Write down shadows you notice. Always include the time, where the sun was and how long or short the shadow was.

Day	Time	What made the shadow	Which way it pointed	Length or size

Questions I still have

Write a question about light or shadows that this book made you wonder about. Then write how you might find out.

Answers

Try every puzzle first. Where an answer shows a calculation, check the steps, not just the result.

Chapter 1

Multix on the corner. Staring cannot bend light. Light from behind the corner travels in straight lines and is blocked by the wall. **Three cards.** A dot shows when the holes line up. It disappears when one card moves. **Keep box.** Kept: the direction light goes. Set aside: its colour and brightness.

1.1 (a) One shadow, if the torch is exactly in line. (b) Each pencil blocks the same line of light, so the shadows fall on top of each other.
1.2 (a) S (b) N, a reflection is light bouncing, not blocked. (c) S (d) N.
1.3 (a) A straight line from the torch through the hole to the wall. (b) Up. The dot moves 10 cm up, twice the card's move, because the wall is twice as far as the card.
1.5 (a) $8 \times 10 \div 40 = 2$ cm. (b) Bigger: the nearer the object, the bigger the picture.
1.4 Lines of light from the ball to your eye run through the wall and are blocked.

Chapter 2

Multix on bigger shadows. No. Moving the wall as well as the puppet changes two things, so the test cannot say which made the shadow bigger. **Keep box.** Kept: the lines of light and the heights. Set aside: the fuzzy edges of the shadow.

2.1 (a) $10 \times 60 \div 15 = 40$ cm. (b) $10 \times 60 \div 60 = 10$ cm.
2.2 (a) The distance from the puppet to the torch. (b) The height of the torch. (c) Keep the torch in place and change only the puppet's distance.
2.3 (a) 12 cm. (b) 1 cm. (c) A 1 cm miss is about the size of the other misses. Measure again before changing your trust.
2.4 (a) B. (b) S. (c) B.
2.5 (a) $18 \times 60 \div 20 = 54$ cm. (b) $18 \times 60 \div 40 = 27$ cm.
2.6 20 cm and 40 cm. The second is twice as tall.
2.7 $10 \times 80 \div 20 = 40$ cm. It grew, from 30 cm.
2.8 About 20 cm. About 40 cm.
2.9 Light travels in straight lines from the torch past the hand to the wall. A hand nearer the torch blocks a wider spread of lines, so the lines are steeper and land higher and wider on the wall.

Chapter 3

Multix on noon. No. The sun is highest at solar noon, which is not always 12:00 on the clock. **Keep box.** Kept: the direction and height of the sun, shown as a shadow. Set aside: its brightness and colour.

3.1 (a) 12:00. (b) $45 - 26 = 19$ cm. (c) The 14:00 shadow, 35 cm against 34 cm.
3.2 West. At 3:00 in the afternoon in Canada the shadow points east.
3.3 (a) M (b) N (c) A.
3.4 The sun is south. A high sun makes a short shadow.

3.6 Yes. The shadows are almost equal in length (45 and 46 cm), so the halfway line points north.
3.7 No. The lengths are very different, so the two tips are not at matching times either side of noon.
3.8 December. A long noon shadow means the sun is low, which happens in winter.
3.9 $76 \div 30 \approx 2.5$, so about 3 when rounded to the nearest whole number.
3.5 $22 \div 20 = 1.1$, so 40 cm gives 44 cm and 10 cm gives 11 cm.

Chapter 4

Multix on the dark. No. A clock only works if the dots mark where the sun's shadow really falls at each hour. **Keep box.** Kept: the shadow's place at marked hours. Set aside: minutes between marks and the season.

4.1 11:30. The friend is wrong: a long shadow could be morning or afternoon. The place of the tip is what tells the time.
4.2 (a) 6, 9 and 3 min. (b) $18 \div 3 = 6$ min. (c) $9 - 3 = 6$ min.
4.3 (a) $12 - 4 = 8$ min. (b) Yes, 8 is bigger than 2 and 3. (c) The clock drifts as the weeks pass.
4.4 Question: does the shadow clock still agree with a watch? Change: the day. Measure: minutes between shadow clock and watch. Keep the same: the stick, the board, the place, the hours checked.
4.5 Clouds block the sun, so there is no shadow. People used other tools, such as a water clock or a marked candle.

Chapter 5

Multix on angles. No. Measure again and check the protractor. A 30 degree beam leaves at 30, not 60. **Keep box.** Kept: the angle in and angle out. Set aside: the width of the beam and the surface's bumps.

5.1 Maze 1: F (3 mirror bounces). Maze 2: N (5 bounces). Maze 3: K (6 bounces).
5.2 North hitting (/) goes east. West hitting (\) goes north.
5.3 Two parallel mirrors, each at 45 degrees, one at the top of the periscope and one at the bottom.
5.5 MOM, WOW and TOOT.
5.6 9:00. The mirror swaps left and right, so the hour hand moves to the other side. Reasons for the second part vary; any careful reason that mentions the mirror flipping the picture left to right is fine.
5.7 Any maze with three mirrors, checked by tracing.
5.8 Maze A leaves at R, with 5 bounces. Maze B leaves at D, with 5 bounces.
5.9 Yes. Light paths can be run in reverse. Sent in at F, the beam of Maze 1 comes out where the first beam entered.
5.4 40 degrees. Off by 4 degrees. That is a bit more than the 1 to 2 degrees in the practice runs, so measure again.

Chapter 6

Multix on the model. No. A model keeps some things and drops others. A shadow clock drops the minutes, the season and the weather.

6.1 (a) S (b) B (c) G.
6.2 (a) M (b) R (c) M.
6.4 8:00 L, 10:00 M, 12:00 S, 14:00 M, 16:00 L.
6.5 (a) The slope of the ground and the sticks' place. (b) Two things changed, so a bigger shadow cannot be traced to one.
6.6 The building's shadow is 10 times the stick's shadow, so the building is 10 times as tall: 300 cm.
6.7 Keeps the direction of the shadow, drops the minutes or the season.
6.3 Examples: the colour of the sun, the minutes, the season, the weather.
Slop check. (a) You cannot tell: no title, no axes, no labels. (b) Many colours, stars, joke text, tilted filler. (c) A title, the hours, the shadow lengths and units. (d) The stars and jokes.
Redraw. One title, labelled axes with hours and centimetres, calm colours.

Whole-book table. Chapter 2: simplified a shadow to a line drawing, kept the straight lines and the heights, set aside fuzziness. Chapter 3: simplified the sun to a direction and a height, kept the shadow length and direction, set aside brightness. Chapter 4: simplified the day to marked dots, kept the hour dots, set aside the minutes and the season. Chapter 5: simplified a mirror to a rule, kept the angles, set aside bumps. Chapter 6: simplified a project to a report, kept the pattern, set aside the detail of each step.

Words

gnomon the stick or edge on a sundial whose shadow tells the time.

model a simple stand-in that keeps some things and drops others.

normal a line at a right angle to a mirror at the point where light hits.

parallel lines that run side by side and never meet.

ratio how many times one number holds another, such as shadow divided by height.

reflect to bounce off a surface.

shadow the place where light is blocked.

solar noon the time when the sun is highest, and the shadow shortest.

sundial a clock that uses a shadow.

Notes for grown-ups

This book teaches that a shadow, a drawing and a clock are models: simple stand-ins for light and the sun. Each chapter has one activity, one worked example with made-up practice numbers, one trust box, one "what did we keep?" prompt and a page of puzzles.

Materials and safety. A torch, a stick, card, a mirror and a sunny day. Never look at the sun, and never aim a mirror at eyes or use sunlight on one. Keep the torch low. The shadow tracing is done on a flat, safe spot outdoors.

Directions. The shadow directions in Chapter 3 are for the northern half of the world. South of the equator, the midday shadow points south.

The practice numbers were made up so the arithmetic is easy to check. Real shadows depend on the place, the day of the year and the clock, so real results will differ.

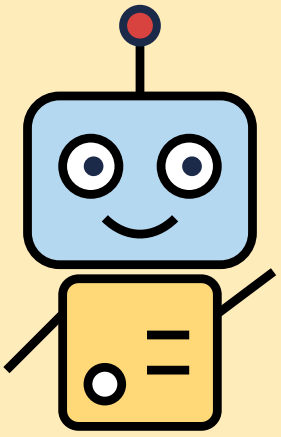
Sources

- E. Hecht. *Optics*, 5th edition. Pearson, 2016. For the law of reflection and straight-line light.
- A. E. Waugh. *Sundials: Their Theory and Construction*. Dover, 1973. For the gnomon and how sundials work.
- Cleomedes. *On the Circular Motions of the Celestial Bodies*. For the later report of Eratosthenes' shadow measurement.

Next in Science

Things That Push Back

A box slides across carpet and stops sooner than it does on paper. Book 4 shows how to find the pushes and pulls on a thing, draw each as an arrow, and balance a seesaw from a ruler.



Ages 9 to 11 ⋆ Science Book 3